

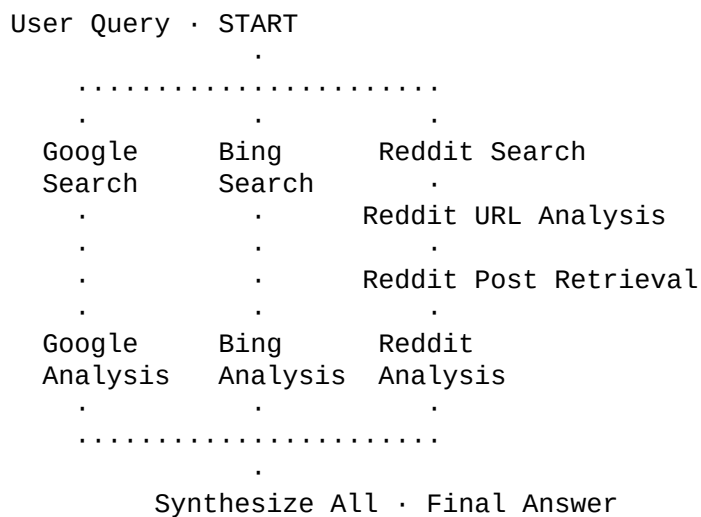
Advanced Langflow Web Agent

A **multi-source research agent** built with LangGraph that searches **Google, Bing, and Reddit** in parallel, analyzes results from each source independently, and synthesizes a comprehensive final answer.

· Features

- **Parallel multi-engine search** · Google, Bing, and Reddit simultaneously
 - **Per-source AI analysis** · Each source gets its own LLM analysis pass
 - **Reddit deep-dive** · Fetches and analyzes actual Reddit post content
 - **Synthesis** · Merges all analyses into one cohesive answer
 - **Bright Data integration** · Uses SERP and web scraping APIs
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.. Architecture



· File Structure

| File | Purpose | |-----|-----| | `main.py` | LangGraph state machine with 9 nodes, parallel branching, and synthesis | | `prompts.py` | `PromptTemplates` class · system/user prompts for each search source | | `web_operations.py` | Bright Data API integration for SERP and Reddit searches | | `snapshot_operations.py` | Async polling and download logic for search results | | `pyproject.toml` | Dependencies and project metadata |

· Getting Started

Prerequisites

- Python 3.10+
- OpenAI API key
- Bright Data API credentials

Installation

```
cd Advanced-Langflow-Web-Agent
uv sync
```

Environment Variables (.env)

```
OPENAI_API_KEY=sk-...
BRIGHT_DATA_API_KEY=...
```

Run

```
uv run python main.py
```

· Logic Flow

- 1 **Input** · User provides a research question
- 2 **Parallel Search** · Three concurrent API calls via Bright Data
- 3 **Reddit Enhancement** · Reddit URLs analyzed and full post content retrieved
- 4 **Individual Analysis** · Each source analyzed independently by GPT
- 5 **Synthesis** · All three analyses combined into one comprehensive answer

Key Functions

Function	File	Description
<code>serp_search()</code>	<code>web_operations.py</code>	Triggers SERP search via Bright Data
<code>reddit_search_api()</code>	<code>web_operations.py</code>	Reddit-specific search
<code>reddit_post_retrieval()</code>	<code>main.py</code>	Fetches full Reddit post content
<code>synthesize_analyses()</code>	<code>main.py</code>	Merges all source analyses

· Dependencies

- `langchain >= 0.3.27` · `langchain-openai >= 0.3.29` · `langgraph >= 0.6.4` · `python-dotenv`
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· License

Educational project · use freely for learning and reference.